



NTNU – Trondheim
Norwegian University of
Science and Technology

TTT4120 Digital Signal Processing Fall 2025

Lecture: Discrete-Time Systems in Time-Domain

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Lecture in course book*

- Proakis, Manolakis Digital Signal Processing, 4th Ed.
 - 2.2 Discrete-time systems
 - 2.3 Analysis of discrete-time linear time-invariant systems
 - 2.4 Recursive and non-recursive discrete-time systems
 - 2.4.2 Linear time-invariant systems characterized by constant-coefficient difference equations
 - 2.5.1 Structures for the realization of linear time-invariant systems

*Level of detail is defined by lectures and problem sets

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Contents and learning outcomes

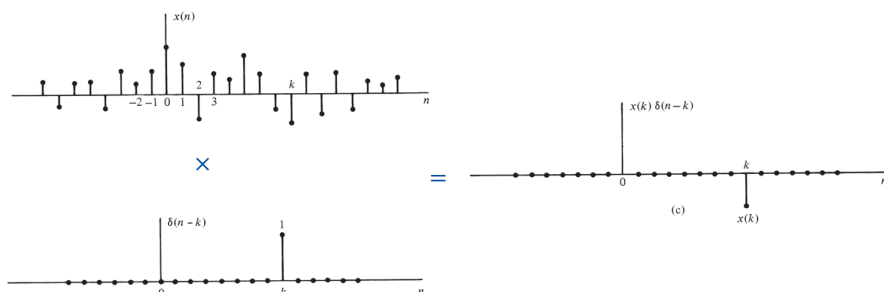
- Signal decomposition using unit impulses
- Discrete-time systems
- Classifications of discrete-time systems
- Linear time-invariant systems and the convolution sum
- Audio demo

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Signal decomposition using unit impulses

- Signal decomposition using sum of delayed unit impulses by exploiting the **sifting property**: $x[k]\delta[n - k] = x[n]\delta[n - k]$



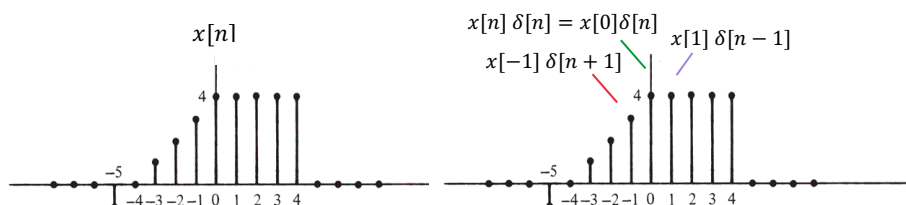
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Signal decomposition using unit impulses...

- Discrete-time signals can be represented by scaled shifted impulses, that is, the impulse shifted by k samples is multiplied by $x[k]$



$$x[n] = \cdots x[-1] \delta[n+1] + x[0] \delta[n] + x[1] \delta[n-1] + x[2] \delta[n-2] + \cdots$$

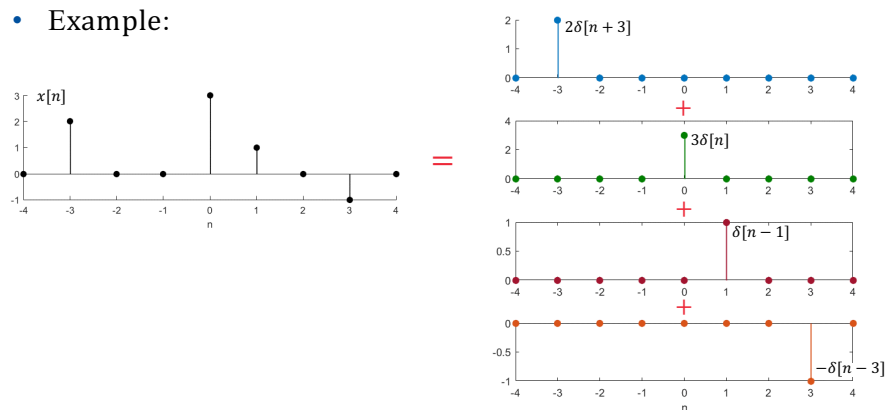
$$= \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

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Signal decomposition using unit impulses...

- Example:



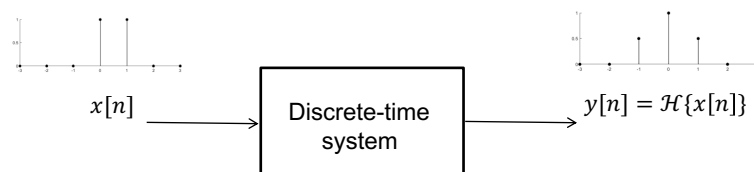
$$x[n] = 2\delta[n+3] + 3\delta[n] + \delta[n-1] - \delta[n-3]$$

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Discrete-time systems

- Discrete-time systems transform (map) an input sequence $x[n]$ to an output sequence $y[n]$



- Mathematically we have

$$y[n] = \mathcal{H}\{x[n]\}$$

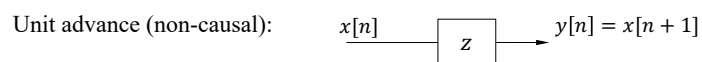
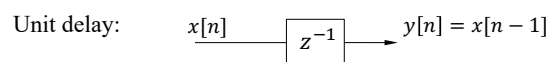
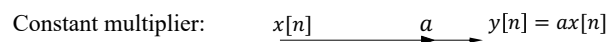
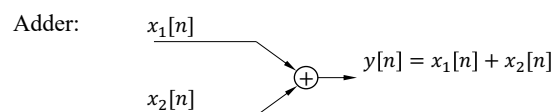
where operator $\mathcal{H}$ describes the discrete-time system

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Discrete-time systems...

- Graphical representation of building blocks



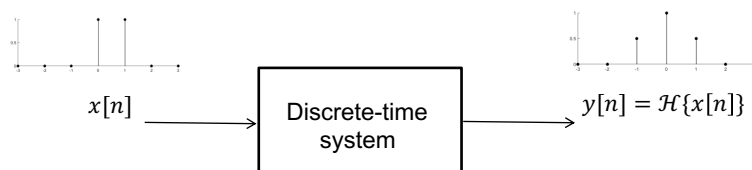
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Classification of discrete-time systems

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Classification of discrete-time systems

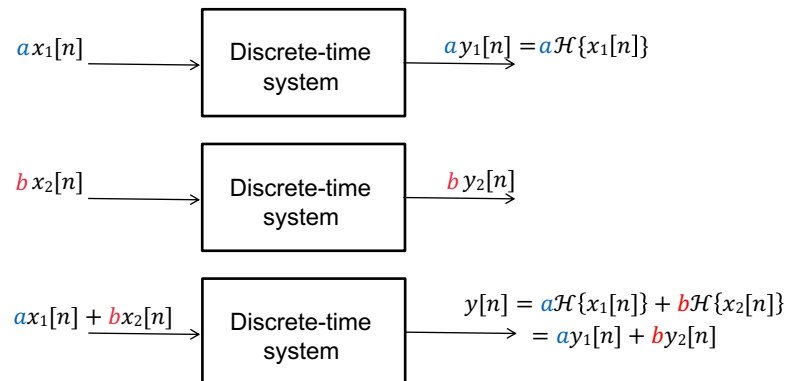


- A discrete-time system can be classified as:
 - linear or nonlinear
 - time invariant or time variant
 - causal or noncausal
- Property must hold for **every possible** input to the system
 - to disprove a property, need a single counter-example
 - to prove a property, need to prove for the general case

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Linear discrete-time systems



- A linear system is a system for which superposition holds

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Linear discrete-time systems...

- Linear (L) or nonlinear (NL) system?

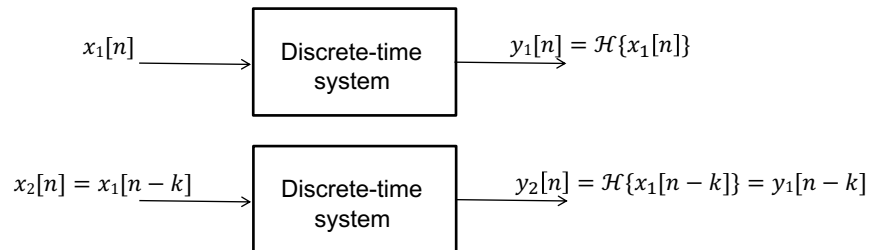
- $y[n] = cx[n]$
- $y[n] = (n + 4)x[n]$
- $y[n] = x[n + 1]$
- $y[n] = x[-n]$
- $y[n] = \sqrt{x[n]} + x^2[n - 2]$
- $y[n] = cx[n] + 3$

Answer: L, L, L, L, NL, NL

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Time-invariant discrete-time systems



- A system whose properties do not vary in time is referred to as being **time invariant**

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Time-invariant discrete-time systems...

- Time-invariant (**TI**) or time-variant (**TV**) system?

1. $y[n] = cx[n]$
2. $y[n] = (n + 4)x[n]$
3. $y[n] = x[n + 1]$
4. $y[n] = x[-n]$
5. $y[n] = \sqrt{x[n]} + x^2[n - 2]$
6. $y[n] = cx[n] + 3$

Answer: **TI**, **TV**, **TI**, **TV**, **TI**, **TI**

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Causal versus noncausal systems

- **Causal system**: output of system at any time n depends only on present and past inputs, i.e.,

$$y[n] = f\{x[n], x[n-1], x[n-2], \dots\}, \forall n$$

- Usually, in the case of a discrete-time signal, a noncausal system is not implementable in real time, since future values are unknown
- Noncausal systems are practical for processing of pre-stored values

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Causal versus noncausal systems...

- Causal (C) or noncausal (NC) system?

1. $y[n] = cx[n]$
2. $y[n] = (n+4)x[n]$
3. $y[n] = x[n+1]$
4. $y[n] = x[-n]$
5. $y[n] = \sqrt{x[n]} + x^2[n-2]$
6. $y[n] = cx[n] + 3$

Answer: C, C, NC, NC, C, C

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Stability

- A discrete-time system is stable if and only if, for every bounded input, the output is also bounded
- A system is **bounded-input bounded-output stable (BIBO)** iff

$$|x[n]| \leq M_x < \infty \Rightarrow |y[n]| \leq M_y < \infty \Rightarrow, \forall n$$

- We want our systems to behave in a predictable manner

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Stability...

- Stable (**S**) or unstable (**US**) system?

1. $y[n] = cx[n]$
2. $y[n] = (n + 4)x[n]$
3. $y[n] = x[n + 1]$
4. $y[n] = x[-n]$
5. $y[n] = \sqrt{x[n]} + x^2[n - 2]$
6. $y[n] = cx[n] + 3$

Answer: **S**, **US**, **S**, **S**, **S**, **S**

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Linear time-invariant system

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Linear time-invariant systems

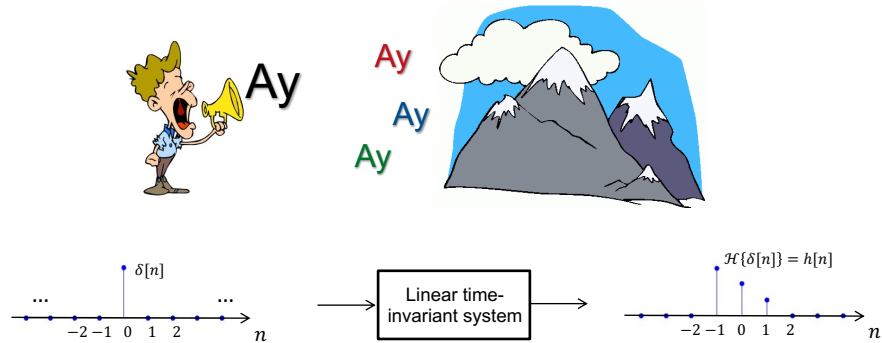
- This course is mostly dealing with linear time-invariant systems
- Knowing the system response to a unit impulse (impulse response), we can calculate the system output for an arbitrary input signal

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Impulse response

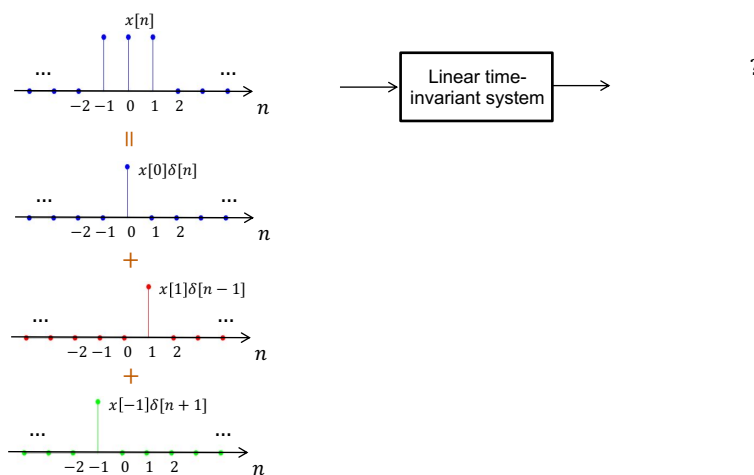
- Send a short impulse into the system and observe the output



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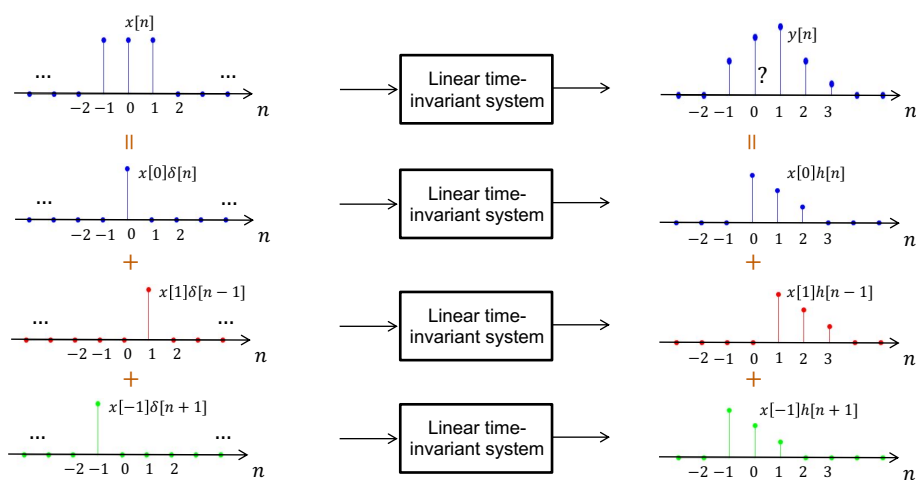
Impulse response and convolution sum



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Impulse response and convolution sum

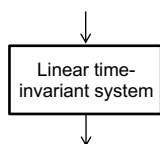


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Impulse response and convolution sum...

$$x[n] = \dots x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + \dots$$



$$y[n] = \dots + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + \dots$$

$$= \sum_k x[k]h[n-k]$$

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Convolution sum

- More formally,

$$\begin{aligned}
 y[n] &= \mathcal{H}\{x[n]\} = \mathcal{H}\left\{\sum_k x[k]\delta[n-k]\right\} \\
 &= \sum_k x[k]\mathcal{H}\{\delta[n-k]\} \\
 &= \sum_k x[k]h[n-k] = x[n] * h[n]
 \end{aligned}$$

- The output of an LTI system is obtained by **convolving** (the asterisk operation) its *impulse response* with the *input signal*

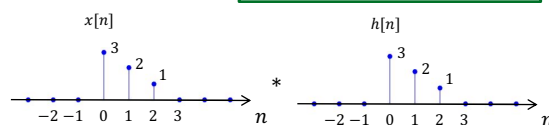
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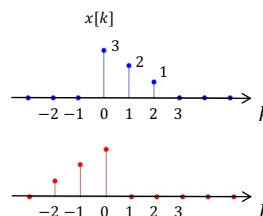
Example: Flip-shift-multiply-sum

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[0] = \sum_k x[k]h[-k]$$



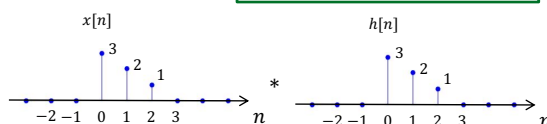
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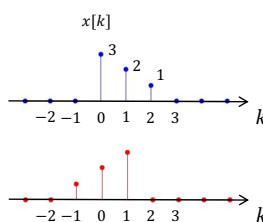
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[1] = \sum_k x[k]h[1-k]$$



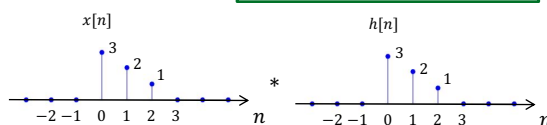
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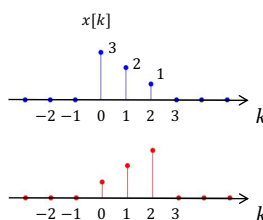
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[2] = \sum_k x[k]h[2-k]$$



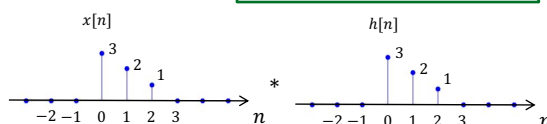
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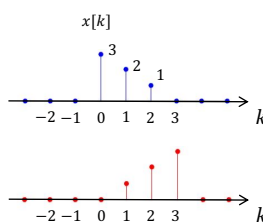
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[3] = \sum_k x[k]h[3-k]$$



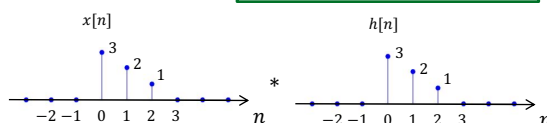
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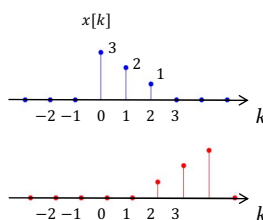
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[4] = \sum_k x[k]h[4-k]$$



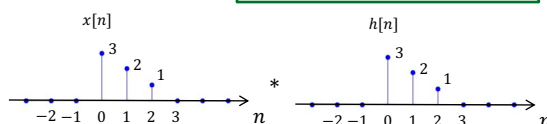
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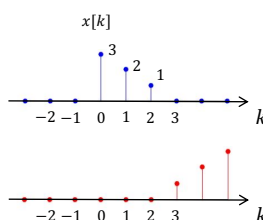
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[5] = \sum_k x[k]h[5-k]$$



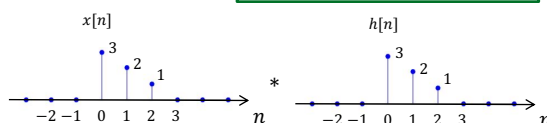
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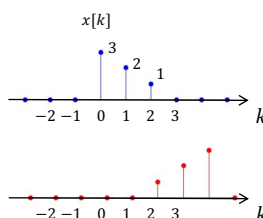
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[4] = \sum_k x[k]h[4-k]$$



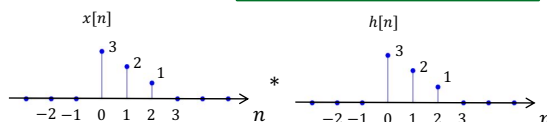
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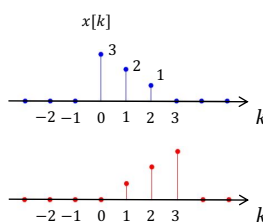
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[3] = \sum_k x[k]h[3-k]$$



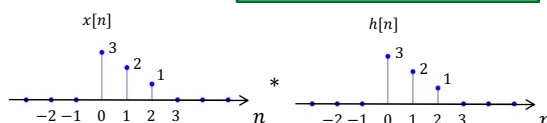
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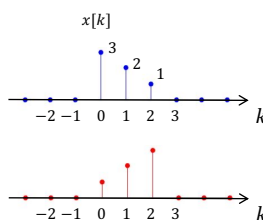
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[2] = \sum_k x[k]h[2-k]$$



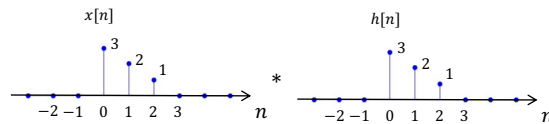
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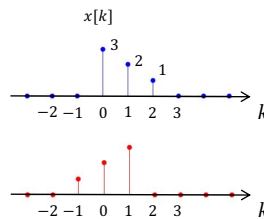
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[1] = \sum_k x[k]h[1-k]$$



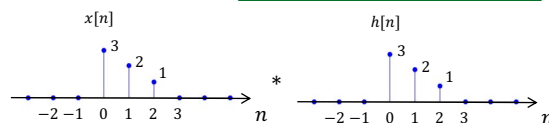
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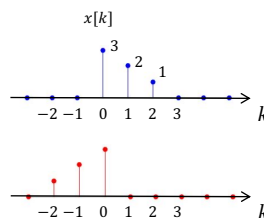
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[0] = \sum_k x[k]h[-k]$$



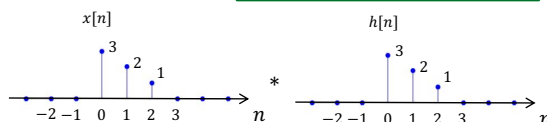
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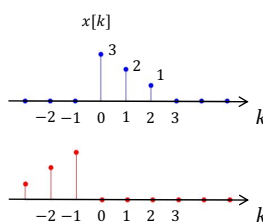
Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



$$y[-1] = \sum_k x[k]h[-1-k]$$



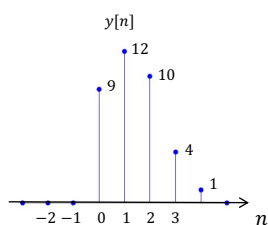
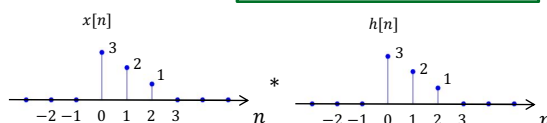
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Example: Flip-shift-multiply-sum...

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



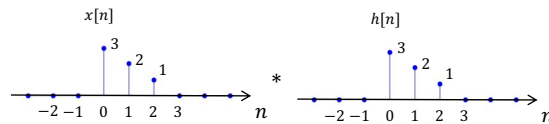
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Example: the easier way

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



- Convolution matrix: multiply and sum anti-diagonals

$x[n] \backslash h[n]$	3	2	1
3	9	6	3
2	6	4	2
1	3	2	1

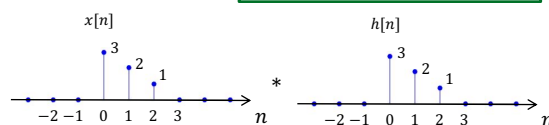
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Example: the easiest way

- What is the output?

$$y[n] = \sum_k x[k]h[n-k]$$



- Let the computer do the job

```
Matlab
x = [3 2 1];
h = [3 2 1];
y = conv(x,h)
```

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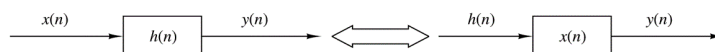
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Properties of convolution

- Commutative:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} x[n-k]h[k] = h[n] * x[n]$$



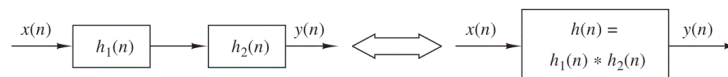
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Properties of convolution...

- Associative:

$$y[n] = (x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$



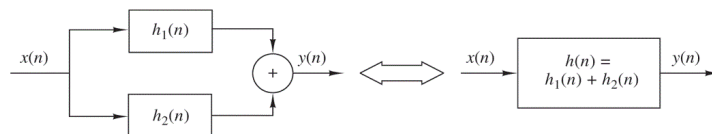
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Properties of convolution...

- Distributive:

$$y[n] = x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$



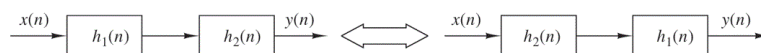
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Properties of convolution...

- Properties can be exploited to change order of building blocks
- Order does not matter!

$$y[n] = (x[n] * h_1[n]) * h_2[n] = (x[n] * h_2[n]) * h_1[n]$$



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Finite length sequences

- If $x[n]$ has finite length N_x and $h[n]$ has finite length N_h
 $\Rightarrow y[n]$ has length $N_y = N_x + N_h - 1$

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=0}^{N_x-1} x[k]h[n-k] \\ &= \{l = n - k\} = \sum_{l=n-N_x+1}^n x[n-l]h[l] \end{aligned}$$

- We have $y[n] = 0$ for $n < 0$ and $n - N_x + 1 \geq N_h$

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Causal linear time-invariant systems

- Output should depend only on past and current inputs

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} h[k]x[n-k] \\ &= \sum_{k=-\infty}^{-1} h[k]x[n-k] + \sum_{k=0}^{\infty} h[k]x[n-k] \end{aligned}$$

- Thus, we must have $h[n] = 0, n < 0$, for causal systems

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Stability of linear time-invariant systems

- Input $x[n]$ is bounded: $|x[n]| \leq M_x < \infty$
- A bounded input $x[n]$ to a linear time-invariant system yields a bounded output $y[n]$, $|y[n]| \leq M_y < \infty$ if $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$$|y[n]| = |\sum_{k=-\infty}^{\infty} h[k]x[n-k]| \leq M_x \sum_{k=-\infty}^{\infty} |h[k]|$$

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FIR and IIR systems

- Infinite(-duration) impulse response (IIR) system is a system whose impulse response $h[n]$ has infinite support

$$y[n] = \sum_{k=0}^{\infty} h[k]x[n-k] \text{ (causal IIR)}$$

- Finite(-duration) impulse response (FIR) system is a system whose impulse response $h[n]$ has finite length

$$y[n] = \sum_{k=0}^{N_h-1} h[k]x[n-k] \text{ (causal FIR)}$$

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Systems described by difference equations

- Characterizing a system using impulse response not always feasible
- An important class of linear time-invariant (LTI) systems can be described by constant-coefficient (real-valued) difference equations

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

- Usually normalized with a_0 , i.e., setting $a_0 = 1$

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

- Special case of FIR when $a_k = 0, k \geq 1$ and $h[n] = b_n, 0 \leq n \leq M$

Matlab

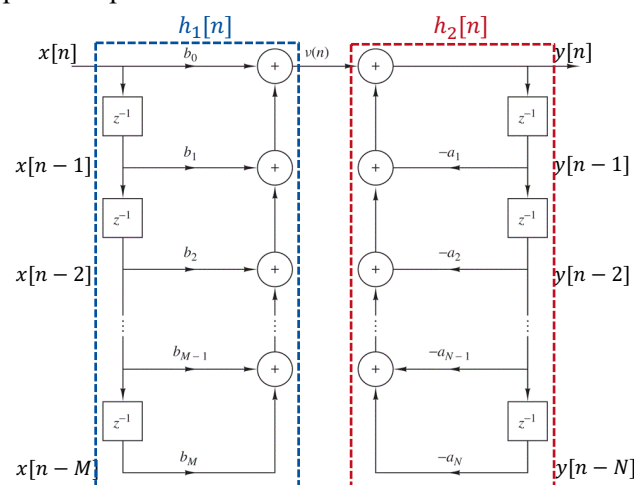
```
b = [b0, b1, ..., bM];  
a = [a0, a1, ..., aN];  
y = filter(b, a, x)
```

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Systems described by difference...

- Graphical representation: Direct form I structure

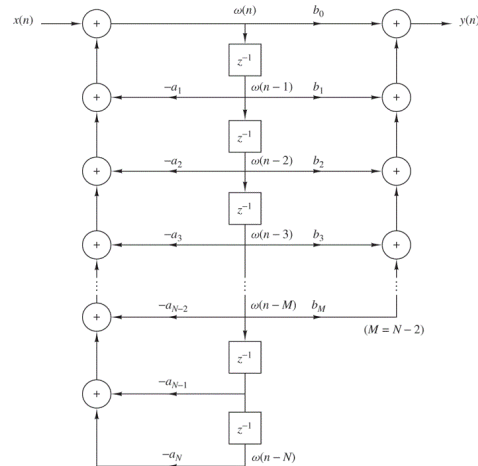


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Systems described by difference...

- Graphical representation: Direct form II structure



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Systems described by difference...

- How to obtain the impulse response $y[n]$ from a difference equation?

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

- Set $x[n] = \delta[n]$ which gives $y[n] = h[n]$

$$h[n] = \sum_{k=0}^M b_k \delta[n-k] - \sum_{k=1}^N a_k h[n-k]$$

$$= b_n - \sum_{k=1}^N a_k h[n-k]$$

- Solve for $h[n]$ sequentially for $n = 0, 1, 2, \dots$
- Requires initial conditions or given a causal system
- Not necessarily closed-form expression

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Systems described by difference...

- General solution can be obtained (see lecture notes)
- Simpler approach is to use transform methods (later)

Matlab

```
b = [b0, b1, ..., bM];  
a = [a0, a1, ..., aN];  
h = impz(b, a, n)
```

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Summary

Today:

- Signal decomposition using delayed unit impulses
- Discrete-time systems and classifications
- Linear time-invariant systems

Next:

- Discrete-time Fourier transform

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